

Pure

Algebra

Polynomials : A polynomial is an expression involving powers of 'x' that are whole number only.

eg. $7x^4 + 2x^2 - 3$

Annotations:
- **coefficient**: points to 7
- **Power**: points to 4
- **Variable**: points to x

0, 1, 2, 3, ...

Expression	polynomial
$49x^{103}$	Yes
$17x^{-2}$	No
$6x^{3.2}$	No
$\frac{1}{x^3}$	No
$\sqrt[3]{x}$	No

subtr.

Mult.

Adding / Subtract / Multiply polynomials

1. $2x^2 + 5x - 7x^2 = -5x^2 + 5x$

2. $2x^2 \times 3x^3 = 6x^5$

3. $(x^2 + 2)(3 - x^3) = 3x^2 - x^5 + 6 - 2x^3$
 $= -x^5 - 2x^3 + 3x^2 + 6$

Division of Polynomials

$$\begin{array}{r}
 141 \rightarrow \text{Quotient} \\
 3 \overline{) 425} \\
 \underline{3} \\
 12 \\
 \underline{12} \\
 05 \\
 \underline{3} \\
 2 \rightarrow \text{Remainder}
 \end{array}$$

$$425 = 3 \times 141 + 2$$

the degree of the quotient is 2

$$\begin{array}{r}
 x^2 + 3x + 2 \\
 x - 2 \overline{) x^3 + x^2 - 4x + 2} \\
 \underline{x^3 - 2x^2} \\
 3x^2 - 4x \\
 \underline{3x^2 - 6x} \\
 2x + 2 \\
 \underline{2x - 4} \\
 6
 \end{array}$$

the degree of the remainder is 0

$$= (x-2) \times (x^2 + 3x + 2) + 6$$

⇒ The degree of a polynomial is the value of the highest power of x

e.g. $x^2 + x - 4$ has degree of 2
 7 has degree of 0

NOTE: Each power of x must be represented in the division
 if a power is missing, add it with a coefficient of "0"

$$\begin{array}{r}
 2x + 1 \overline{) 6x^3 + x - 2} \\
 \text{change to}
 \end{array}$$

$$2x + 1 \overline{) 6x^3 + 0x^2 + x - 2}$$

$$\begin{array}{r}
 3x^2 - \frac{3}{2}x + \frac{5}{4} \\
 2x+1 \overline{) 6x^3 + 0x^2 + x - 2} \\
 \underline{6x^3 + 3x^2} \\
 -3x^2 + x \\
 \underline{-3x^2 - \frac{3}{2}x} \\
 \frac{5}{2}x - 2 \\
 \underline{ \frac{5}{2}x + \frac{5}{4}} \\
 \phantom{\frac{5}{2}x} - \frac{13}{4}
 \end{array}$$

Long Division

Quotient
 Divisor $\overline{)}$ Dividend

Remainder

Degree of Dividend = Degree of Quotient + Degree of Divisor

Degree of Remainder < degree of divisor

NOTE: The previous method is called LONG DIVISION

An alternative method is Equating Coefficients

Example

Divide $x^3 + x^2 - 4x + 2$ by $x - 2$

Answer

$$\text{Let } x^3 + \boxed{1}x^2 - \boxed{4}x + \boxed{2} = (x - 2)(x^2 + Ax + B) + R$$

degree = 2

degree = 0

Equating Coefficients

$$= x^3 + Ax^2 + Bx - 2x^2 - 2Ax - 2B + R$$

$$= x^3 + \boxed{(A-2)}x^2 + \boxed{(B-2A)}x + \boxed{(-2B+R)}$$

So • $A - 2 = 1$

and

• $B - 2A = -4$

$$\boxed{A = 3}$$

$$B - 2(3) = -4$$

$$B - 6 = -4$$

$$\boxed{B = 2}$$

and

• $2 = -2B + R$

$$2 = -2(2) + R \quad \therefore \text{Quotient is } x^2 + 3x + 2$$

$$2 = -4 + R \quad \text{Remainder is } 6$$

$$\boxed{R = 6}$$

eg. Find Quotient and Remainder

When $x^3 - x^2 - x + 12$ is divided by $x + 2$

$$\text{let } x^3 - x^2 - x + 12 = (x + 2)(x^2 + Ax + B) + R$$

$$= x^3 + Ax^2 + Bx + 2x^2 + 2Ax + 2B + R$$

$$= x^3 + (A+2)x^2 + (B+2A)x + (2B+R)$$

So • $A + 2 = -1$ and • $B + 2A = -1$ and $2B + R = 12$

$$A = -3$$

$$B - 6 = -1$$

$$12 + R = 12$$

$$B = 5$$

$$R = 0$$

• Quotient = $x^2 - 3x + 5$

Remainder = 0

OR

$x^3 - x^2 - x + 12$ is divided by $x + 2$
Find remainder

$$x + 2 = 0$$

$$x = -2$$

$$R = (-2)^3 - (-2)^2 - (-2) + 12 = 2$$

The Remainder Theorem (For finding the value of the remainder)

If a polynomial $f(x)$ is divided by a Linear term $(ax+b)$,

Then the remainder is the value of $f\left(-\frac{b}{a}\right)$

$$\text{If } ax+b=0$$

$$ax = -b$$

$$x = -\frac{b}{a}$$

$$\begin{array}{r} \text{Q} \\ x+b \overline{) f(x)} \end{array}$$

R

$$f(x) = (x+b)(Q) + R$$

$$f(-b) = (-b+b)(Q) + R$$

$$f(-b) = R$$

e.g Find the remainder when $2x^3 - x^2 - 3x - 7$ is divided by $2x + 1$

$$2x + 1 = 0$$

$$x = -\frac{1}{2}$$

$$\text{let } f(x) = 2x^3 - x^2 - 3x - 7$$

$$R = f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 - \left(-\frac{1}{2}\right)^2 - 3\left(-\frac{1}{2}\right) - 7 = -\frac{6}{1}$$

The Factor Theorem

If $f(x)$ is a polynomial

and $f(a) = 0$

then $(x-a)$ is a factor of $f(x)$

("a" is the root of the polynomial)

Example Use the factor theorem to factorise

$$f(x) = x^3 - 4x^2 - 7x + 10$$

$$= (x-5)(x+2)(x-1)$$

Answer $f(x) = (x - \underline{a})(x - \underline{b})(x - \underline{c})$

Guess: Try $x = 1$

$$\begin{aligned} f(1) &= (1)^3 - 4(1)^2 - 7(1) + 10 \\ &= 0 \end{aligned}$$

So $(x-1)$ is a factor

→ guess the number to substitute into x and get 0

Divide $f(x)$ by $(x-1)$

$$\begin{aligned} \text{So, } f(x) &= (x-1)(x^2 - 3x - 10) \\ &= (x-1)(x-5)(x+2) \end{aligned}$$

↪ factors

$$\begin{array}{r} x^2 - 3x - 10 \\ x-1 \overline{) x^3 - 4x^2 - 7x + 10} \\ \underline{x^3 - x^2} \\ -3x^2 - 7x \\ \underline{-3x^2 + 3x} \\ -10x + 10 \\ \underline{-10x + 10} \\ 0 \end{array}$$

$$x = 1, x = 5, x = -2 \rightarrow \text{Roots}$$

$x^3 - 4x^2 - 7x + 10$ is exactly divisible by $x-1$

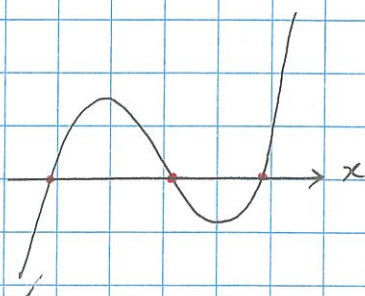
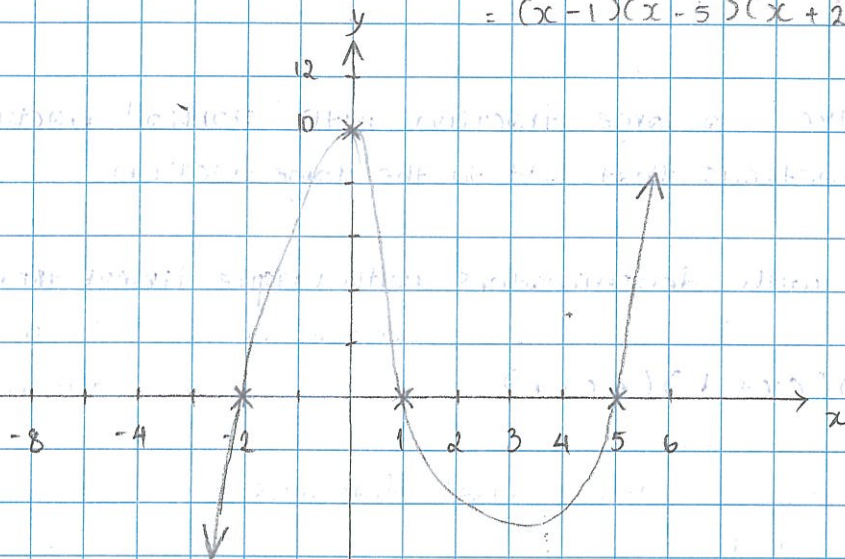
Now, sketch the graph of $f(x) = x^3 - 4x^2 - 7x + 10$.

$$= (x-1)(x-5)(x+2)$$

$+x^2$ Parabola
Symmetrical

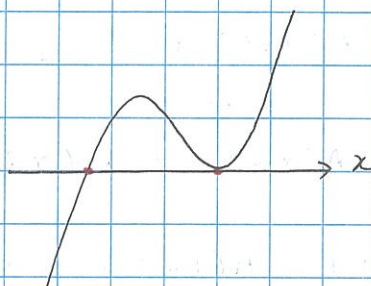
$-x^2$

$+x^3$ or
not symmetrical
 $-x^3$ or
cubic



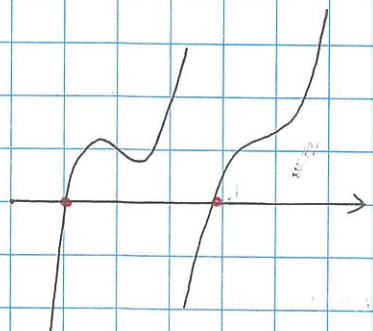
3 real roots

() () ()



2 real roots

() ()²



1 real root

() (quad)

Partial Fractions

$$\frac{3-x}{(x+1)(x-1)} = ? + ?$$

When we "resolve" a large fraction into "partial fractions" we find smaller fractions that add to the large fraction.

Type 1 Fractions with denominators with unique linear factors.
denominators - all linear
all different

$$(ax+b)(cx+d)(ex+f)$$

Eg. Resolve $\frac{3-x}{(x+1)(x-1)}$ into partial fractions

Ans:

memorise

$$\text{Let } \frac{3-x}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$$

cross

Multiply

$$\begin{aligned} 3-x &= \frac{A(x+1)(x-1)}{x+1} + \frac{B(x+1)(x-1)}{x-1} \\ 3-x &= A(x-1) + B(x+1) \end{aligned}$$

Now let $x = 1$

2nd substitution

A omitted

$$3-(1) = A(1-1) + B(1+1)$$

$$2 = 2B$$

$$B = 1$$

let $x = -1$

$$3-(-1) = A(-1-1) + B(-1+1)$$

$$4 = -2A$$

$$A = -2$$

So partial fractions are $\frac{-2}{x+1} + \frac{1}{x-1}$

eg

$$\frac{12x}{(x+1)(2x+3)(x-3)}$$

$$= \frac{A}{x+1} + \frac{B}{2x+3} + \frac{C}{x-3}$$

$$12x = A(2x+3)(x-3) + B(x+1)(x-3) + C(x+1)(2x+3)$$

$$x-3=0$$

$$\bullet \text{ Let } x = 3$$

$$12(3) = A(2(3)+3)(3-3) + B(3+1)(3-3) + C(3+1)(2(3)+3)$$

$$36 = 36C$$

$$C = 1$$

$$\bullet \text{ Let } x = -1 \quad x+1=0$$

$$12(-1) = A(2(-1)+3)(-1-3) + B(-1+1)(-1-3) + C(-1+1)(2(-1)+3)$$

$$-12 = -4A$$

$$A = 3$$

$$\bullet \text{ Let } x = -\frac{3}{2} \quad 2x+3=0$$

$$12\left(-\frac{3}{2}\right) = A\left(2\left(-\frac{3}{2}\right)+3\right)\left(-\frac{3}{2}-3\right) + B\left(-\frac{3}{2}+1\right)\left(-\frac{3}{2}-3\right) + C\left(-\frac{3}{2}+1\right)\left(2\left(-\frac{3}{2}\right)+3\right)$$

$$-18 = \frac{9}{4}B$$

$$B = -8$$

$$\text{So, } f(x) = \frac{3}{x+1} - \frac{8}{2x+3} + \frac{1}{x-3}$$

Type 2 : Denominator contains a quadratic that can't be factorised

$$\frac{3}{(x-1)(x^2+x+1)}$$

$$f(x) = \frac{3}{(x-1)(x^2+x+1)}$$

memorise

$$\text{Let } \frac{3}{(x-1)(x^2+x+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$$

Now use same process as type 1

cross multiply

$$3 = A(x^2+x+1) + (Bx+C)(x-1)$$

$$\text{Let } x = 1 \Rightarrow 3 = A(1^2+1+1)$$

$$3 = 3A$$

$$A = 1$$

$$\text{Let } x = 0 \quad \begin{matrix} \nearrow B \times 0 \end{matrix}$$

$$\Rightarrow 3 = 1(0^2+0+1) + (B(0)+C)(0-1)$$

$$3 = 1 - C$$

$$C = -2$$

now find A, C

using any x

$$\text{Let } x = \text{any number}$$

$$\Rightarrow 3 = 1(2^2+2+1) + (2B-2)(2-1)$$

$$3 = 7 + 2B - 2$$

$$2B = -2$$

$$B = -1$$

$$\text{eg } x = 2$$

$$\text{So, } f(x) = \frac{1}{x-1} + \frac{-x-2}{x^2+x+1}$$

Type 3 Denominator contains a repeated linear factor

$$\frac{1}{(x-a)(x-b)^2}$$

Eg. $\frac{2x-1}{(x+1)(x-2)^2}$

memorise

$$\text{Let } \frac{2x-1}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

Now use same process

cross multiply

$$2x-1 = A(x-2)^2 + B(x+1)(x-2) + C(x+1)$$

$$\begin{aligned} \text{Let } x &= 2 \Rightarrow 2(2)-1 = C(2+1) \\ 3 &= 3C \\ C &= 1 \end{aligned}$$

$$\begin{aligned} \text{Let } x &= -1 \Rightarrow 2(-1)-1 = A(-1-2)^2 \\ -3 &= 9A \\ A &= -\frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{Let } x &= 0 \Rightarrow 2(0)-1 = -\frac{1}{3}(-2)^2 + B(0+1)(0-2) + (1)(0+1) \\ -1 &= -\frac{4}{3} - 2B + 1 \\ -3 &= -4 - 6B + 3 \\ 6B &= 2 \\ B &= \frac{1}{3} \end{aligned}$$

$$\text{So, } f(x) = -\frac{1}{3(x+1)} + \frac{1}{3(x-2)} + \frac{1}{(x-2)^2}$$

Type 4 : Improper Fractions

Degree of Numerator $\geq$ Degree of Denominator.

$$\frac{x^3 - x^2 + 4x - 2}{x^2 - 3x + 2}$$

$$3 > 2$$

or equal

$$2x + 7$$

use long division first

Extra step :

use long division to find Q and R

$$\begin{array}{r} x+2 \\ x^3 - x^2 + 4x - 2 \\ - (x^3 - 3x^2 + 2x) \\ \hline 2x^2 + 2x - 2 \\ 2x^2 - 6x + 4 \\ \hline 8x - 6 \end{array}$$

$$\frac{10}{7} = 1 + \frac{3}{7}$$
$$7 \overline{)10} \begin{array}{r} 1 \\ 7 \\ \hline 3 \end{array}$$

$$\text{So, } \frac{x^3 - x^2 + 4x - 2}{x^2 - 3x + 2} = \frac{x+2}{x^2 - 3x + 2} + \frac{8x - 6}{x^2 - 3x + 2}$$

Now resolve remainder into partial fractions

$$\text{Let } \frac{8x - 6}{x^2 - 3x + 2} = \frac{8x - 6}{(x-2)(x-1)}$$

(type 1)

$$= \frac{A}{x-2} + \frac{B}{x-1}$$

$$8x - 6 = A(x-1) + B(x-2)$$

$$\text{Let } x = 1 \quad 8(1) - 6 = B(1-2)$$

$$2 = -B$$

$$B = -2$$

$$\text{Let } x = 2 \quad 8(2) - 6 = A(2-1)$$

$$10 = A$$

$$\text{SO, } f(x) = x + 2 + \frac{10}{x-2} + \frac{-2}{x-1}$$

$$\text{eg. } \frac{2x+7}{x-1}$$

(type 4)

$$\begin{array}{r} 2 \\ x-1 \overline{) 2x+7} \\ \underline{2x-2} \\ 9 \end{array}$$

$$\text{SO, } \frac{2x+7}{x-1} = 2 + \frac{9}{x-1}$$

STOP HERE

not type 1 or 2 or 3

$$\text{eg } \frac{x^3+2x+1}{x(x^2+1)}$$

$$\frac{x^3+2x+1}{x^3+x}$$

$$\begin{array}{r} 1 \\ x^3+0x^2+x \overline{) x^3+0x^2+2x+1} \\ \underline{x^3+0x^2+x} \\ x+1 \end{array}$$

$$\text{SO, } \frac{x^3+2x+1}{x(x^2+1)} = 1 + \frac{x+1}{x(x^2+1)}$$

Type 2

$$\text{Let } \frac{x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$\text{Let } x+1 = A(x^2+1) + x(Bx+C)$$

$$B+C=0 \quad (1)$$

$$2B+C=-1 \quad (2)$$

$$(2)-(1): B=-1$$

$$(-1)+C=0$$

$$C=1$$

$$\text{Let } x=0$$

$$1=A$$

$$\text{Let } x=1$$

$$2=2+B+C$$

$$B+C=0$$

$$\text{Let } x=2$$

$$3=5+4B+2C$$

$$-2=2(2B+C)$$

$$2B+C=-1$$

$$\text{SO, } f(x) = 1 + \frac{1}{x} + \frac{1-x}{x^2+1}$$

Binomial Expansion

any number

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

has to be one

Formula is only valid if n is a rational number
and $-1 < x < 1$ ($|x| < 1$)

eg. $\left(1 + \frac{1}{2}\right)^{-1} = 1 + (-1)\left(\frac{1}{2}\right) + \frac{(-1)(-2)}{2 \times 1}\left(\frac{1}{2}\right)^2 + \frac{(-1)(-2)(-3)}{3 \times 2 \times 1}\left(\frac{1}{2}\right)^3 + \dots$

$\frac{2}{3}$

$$\frac{(-1)(-2)(-3)(-4)}{4 \times 3 \times 2 \times 1}\left(\frac{1}{2}\right)^4 + \dots$$

$$= 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} + \dots$$

the value is getting smaller.

Example. write $\sqrt{(1+3x)^3}$ in ascending powers of x up to $x^{\frac{3}{2}}$

$$(1+3x)^{\frac{3}{2}} = 1 + \left(\frac{3}{2}\right)(3x) + \frac{\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)}{2 \times 1}(3x)^2 + \frac{\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{3 \times 2 \times 1}(3x)^3 + \dots$$

$$= 1 + \frac{9}{2}x + \frac{\frac{3}{4}}{2}(9x^2) + \frac{-\frac{3}{8}}{6}(27x^3) + \dots$$

$$= 1 + \frac{9}{2}x + \frac{27}{8}x^2 - \frac{81}{48}x^3 + \dots$$

$$= 1 + \frac{9}{2}x + \frac{27}{8}x^2 - \frac{27}{16}x^3 + \dots$$

This is valid if $-1 < 3x < 1$ ie $-\frac{1}{3} < x < \frac{1}{3}$

Expansion of $(a+x)^n$

Eg $(2+x)^{-3}$

take a factor of 2^n out of the bracket.

$$= \left(2 \left(1 + \frac{x}{2} \right) \right)^{-3}$$

$$= 2^{-3} \left(1 + \frac{x}{2} \right)^{-3}$$

$$= \frac{1}{8} \left(1 + (-3) \left(\frac{x}{2} \right) + \frac{(-3)(-4)}{2 \times 1} \left(\frac{x}{2} \right)^2 + \frac{(-3)(-4)(-5)}{3 \times 2 \times 1} \left(\frac{x}{2} \right)^3 + \dots \right)$$

$$= \frac{1}{8} \left(1 - \frac{3}{2}x + \frac{3}{2}x^2 - \frac{5}{4}x^3 + \dots \right)$$

$$= \frac{1}{8} - \frac{3}{16}x + \frac{3}{16}x^2 - \frac{5}{32}x^3 + \dots$$

Partial Fractions combined with Binomial Expansion

Example: Expand $\frac{1+x}{(1-x)(1+x^2)}$ in ascending power of x up to x^5

Method ① split into partial fractions.

② Expand each partial fraction

③ Add the expansion together and simplify

$$\text{Let } \frac{1+x}{(1-x)(1+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x^2}$$

$$(1+x) = A(1+x^2) + (Bx+C)(1-x)$$

Let $x = 1$

$$2 = 2A$$

$$A = 1$$

Let $x = 0$

$$1 = A + C$$

$$C = 1 - 1$$

$$C = 0$$

Let $x = (-1)$

$$0 = 2 + (-B+C)(1+1)$$

$$0 = 2 - 2B$$

$$B = 1$$

$$\text{So, } f(x) = \frac{1}{1-x} + \frac{x}{1+x^2} \quad (1)$$

$$= (1-x)^{-1} + x(1+x^2)^{-1}$$

Now, expand

(2)

$$\begin{aligned} (1-x)^{-1} &= 1 + (-1)(-x) + \frac{(-1)(-2)(-x)^2}{2 \times 1} + \frac{(-1)(-2)(-3)(-x)^3}{3 \times 2 \times 1} + \frac{(-1)(-2)(-3)(-4)(-x)^4}{4 \times 3 \times 2 \times 1} + \frac{(-1)(-2)(-3)(-4)(-5)(-x)^5}{5 \times 4 \times 3 \times 2 \times 1} + \dots \\ &= 1 + x + x^2 + x^3 + x^4 + x^5 + \dots \quad (1) \end{aligned}$$

$$\begin{aligned} x(1+x^2)^{-1} &= x \left(1 + (-1)(x^2) + \frac{(-1)(-2)(x^2)^2}{2 \times 1} - \dots \right) \\ &= x(1 - x^2 + x^4 - \dots) \end{aligned}$$

$$= x - x^3 + x^5 - \dots \quad (2)$$

Now Add:

(1)

(2)

(3)

$$\begin{aligned} \frac{1+x}{(1-x)(1+x^2)} &= \frac{(1-x)^{-1} + x(1+x^2)^{-1}}{(1-x)(1+x^2)} \\ &= \frac{1 + x + x^2 + x^3 + x^4 + x^5 + \dots}{1 - x^2} \\ &= 1 + 2x + x^2 + x^4 + 2x^5 + \dots \end{aligned}$$

$$\text{So, } \frac{1+x}{(1-x)(1+x^2)} = 1 + 2x + x^2 + x^4 + 2x^5 + \dots$$

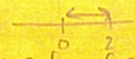
The Modulus Function

$|a|$ = distance of a from 0

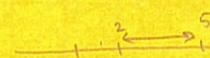
$|a-b|$ = distance of a from b

Distances are always positive

Modulus = Distance

$|2| = 2$  the distance of 2 from 0

$|-3| = 3$ 

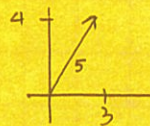
$|5-2| = 3$ 

$|2-5| = 3$

$\left| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right| = 5$

vector

$\sqrt{3^2 + 4^2} = 5$



Modulus Equations

Easy ① $|x| = \text{number}$

$|x| = 3$

$x = +3 \text{ or } -3$

$|x-0| = 3$

Solution

$x = \begin{matrix} + \\ \text{or} \\ - \end{matrix} \text{ number}$

② $|x-a| = b$

→ solve

$x-a = b \text{ or } x-a = -b$

the distance

of x from

$|x-2| = 5$

2 is 5

Solution $x-2 = 5 \text{ or } x-2 = -5$

$x = 7 \text{ or } x = -3$

• $|2x+1| = 4$

$2x+1 = 4$

or

$2x+1 = -4$

$2x = 3$

$2x = -5$

$x = \frac{3}{2}$

$x = -\frac{5}{2}$

$x = 1.5$

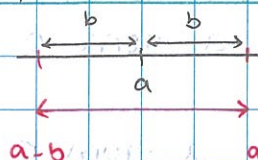
$x = -2.5$

Simple Inequalities

If $|x - a| < b$

then, $-b < x - a < b$

$$a - b < x < b + a$$



Eg. $|x - 1| \leq 23$

$$-23 \leq x - 1 \leq 23$$

$$-22 \leq x \leq 24$$

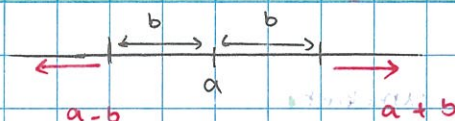
If $|x - a| > b$

solution,

$$x > a + b$$

or

$$x < a - b$$



More difficult equation

$$|x - a| = |x - b|$$

Method ① Square both sides

② Form quadratic = 0

③ Solve quadratic

$$|x - 2| = |2x - 1|$$

$$(x - 2)^2 = (2x - 1)^2$$

$$x^2 - 4x + 4 = 4x^2 - 4x + 1$$

$$3x^2 - 3 = 0$$

$$3(x^2 - 1) = 0$$

$$(x - 1)(x + 1) = 0$$

$$x = 1 \text{ or } x = -1$$

$$|1 - 2| = |2(1) - 1|$$

$$1 = 1$$

$$|-1 - 2| = |2(-1) - 1|$$

$$3 = 3$$

①

②

③

More difficult inequalities

$$|x - a| < |x - b|$$

- Method
- ① square both sides
 - ② quadratic < 0
 - ③ solve quadratic inequality

$$|x - 2| < |2x - 3|$$

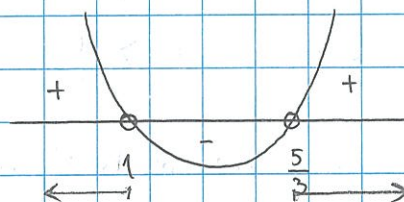
$$(x - 2)^2 < (2x - 3)^2$$

$$x^2 - 4x + 4 < 4x^2 - 12x + 9$$

$$0 < 3x^2 - 8x + 5$$

roots $(3x - 5)(x - 1) = 0$

$$x = \frac{5}{3} \text{ or } 1$$



Solution $x < 1 \text{ or } x > \frac{5}{3}$

14.2.20

Special Examples

outside the modulus $| |$

① Solve $4|x| > |x - 1|$

square both sides including 4

$$(4|x|)^2 > (|x - 1|)^2$$

$$16x^2 > x^2 - 2x + 1$$

$$15x^2 + 2x - 1 > 0$$

etc

$$4|x| = |4x|$$

② only one side has the modulus

$$2x > |x - 1|$$

⇒ use the same method, but check each part of solution

$$(2x)^2 > (x - 1)^2$$

$$4x^2 > x^2 - 2x + 1$$

$$3x^2 + 2x - 1 > 0$$

roots $(3x - 1)(x + 1) = 0$

$$x = \frac{1}{3}, x = -1$$



$$x < -1, x > \frac{1}{3}$$

check $x = -2$

$$2(-2) > |-2 - 1|$$

$$-4 > 3 \quad \text{FALSE}$$

check $x = 1$

$$2(1) > |1 - 1|$$

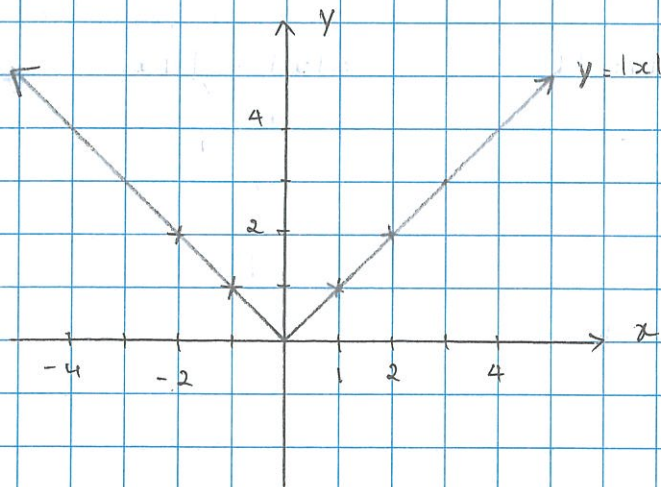
$$2 > 0 \quad \text{TRUE}$$

So, the actual solution is $x > \frac{1}{3}$

Graphing Modulus Functions

① Simplest: $y = |x|$

y value is the modulus of x value

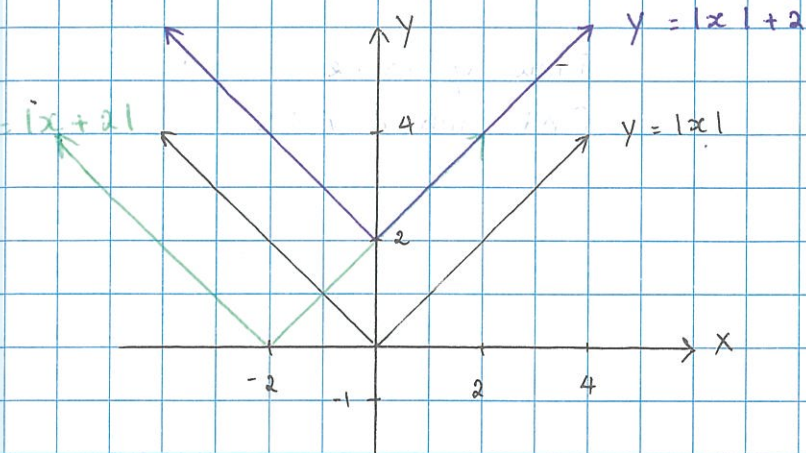


x	y
-2	2
-1	1
0	0
1	1
2	2

② Transformations

$$y = |x| + 2 \quad \text{vertical}$$

$$y = |x + 2| \quad \text{Horizontal}$$

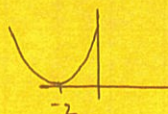


Parabola

$$y = x^2 + 2$$



$$y = (x + 2)^2$$



$$y = -x^2$$

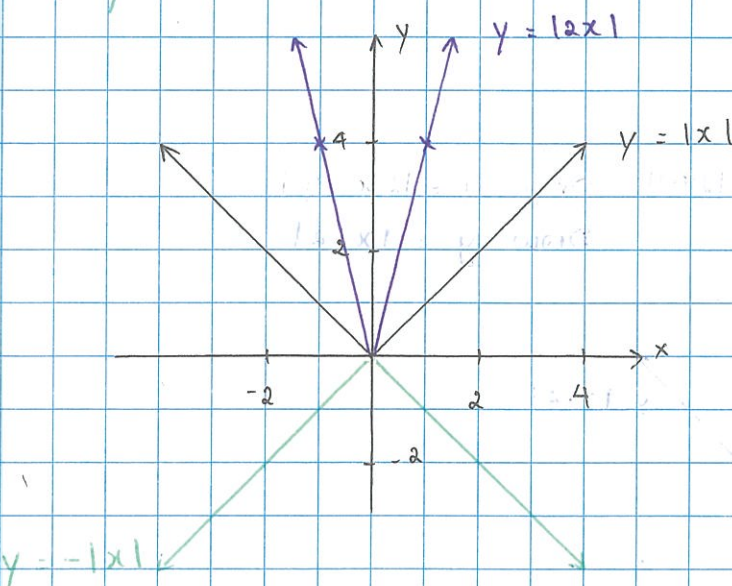


$$y = 2x^2$$

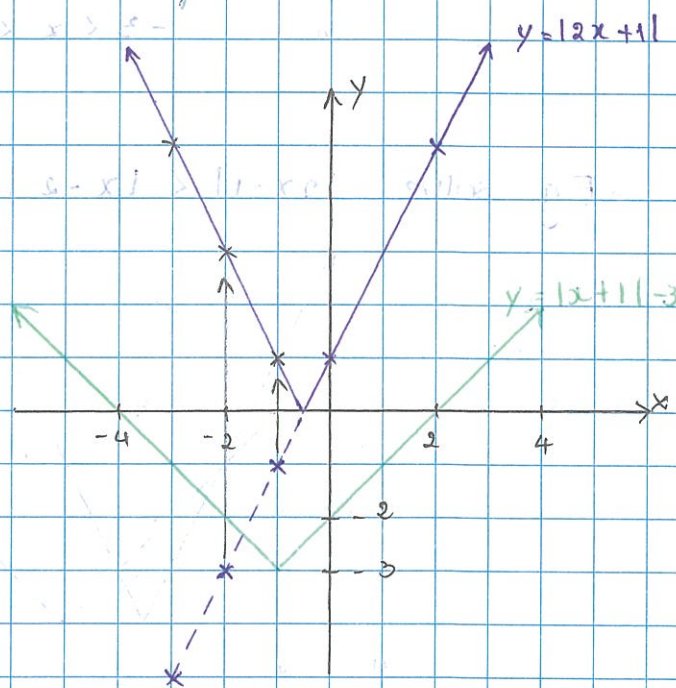


$$y = |2x|$$

$$y = -|x|$$



$$y = |x + 1| - 3$$



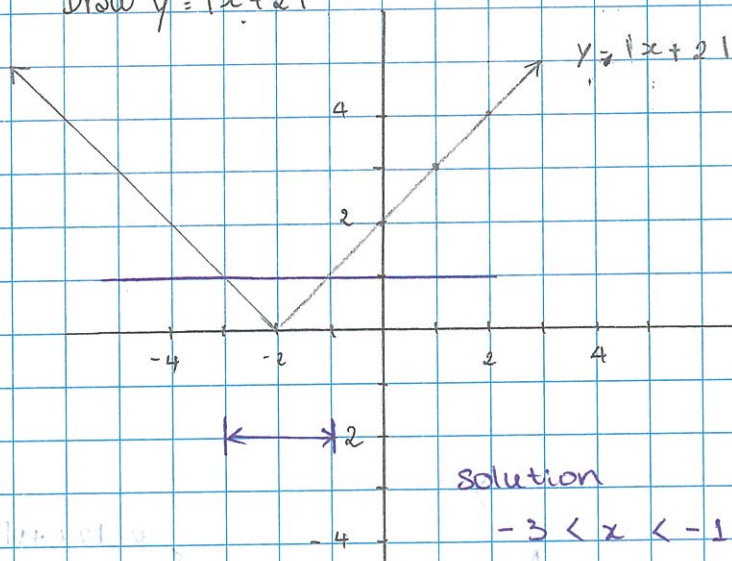
$$y = |2x + 1|$$

Draw $y = 2x + 1$ first
and reflect any negative
y values vertically

Using graphs to solve modulus equations / inequalities

- E.g solve $|x+2| < 1$ (using graph)

Draw $y = |x+2|$



Draw $y = |x+2|$

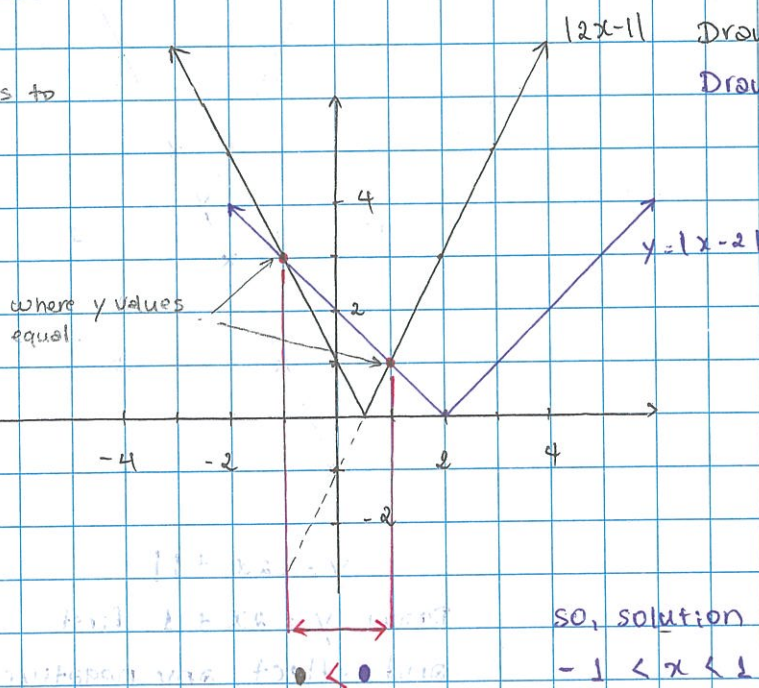
Find where $y < 1$

solution

$$-3 < x < -1$$

- E.g solve $|2x-1| < |x-2|$

(value)
 $y = |2x-1|$ has to
be less than
 $y = |x-2|$



Draw $y = |2x-1|$

Draw $y = |x-2|$

where y values
equal

so, solution

$$-1 < x < 1$$

This method won't give you the
exact value.

• E.g Solve

$$2x < |x-3|$$

Draw

$$y = 2x$$

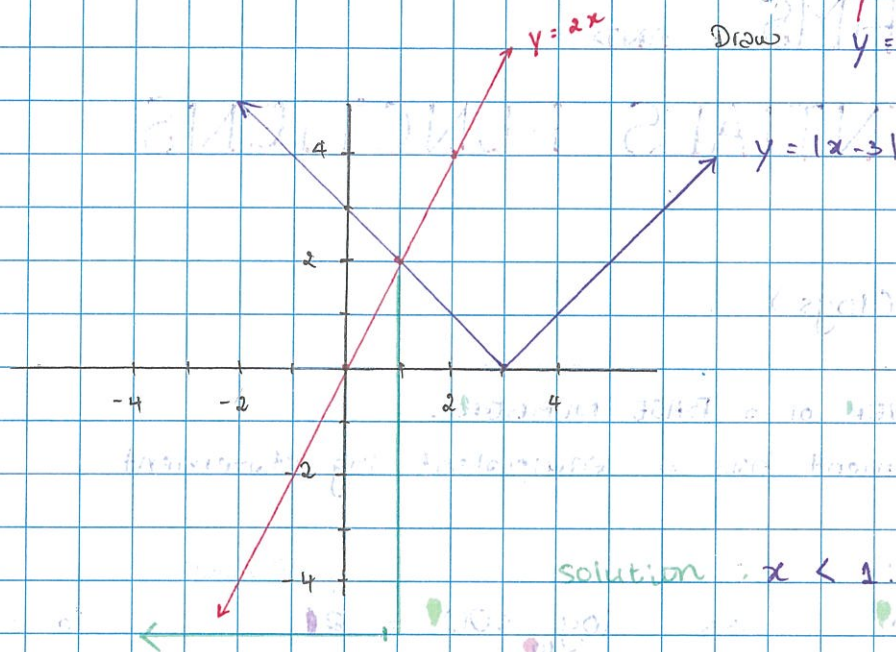
red one has

Draw

$$y = |x-3|$$

to be below

the green one



solution : $x < 1$